

The Attralucian Essays:

Exploring the Finite



First Edition

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The Attralucian Essays



On the
Greene-Pascal Limit

Kevin R. Haylett

Overview

This paper names a limit. It proposes that the sayable has a threshold as well as a floor. The floor has been described elsewhere as the Alphonic Limit — the smallest difference that can be registered as non-single under a given apparatus and set of conditions. The threshold is named here the Greene-Pascal Limit: the region in which a string of symbols begins to cohere, where it can be placed into the Dynamic Semantic Lattice and held as a Functional Symbolic Trajectory. The limit is not a wall. It is a region whose condition is coherence, and coherence needs wider language to hold it. It is conditioned by the lattice, by the licence under which the symbol travels, and by the provenance that survives compression. It moves when the language gets finer. The paper argues that two historical figures — Robert Greene, writing in Cambridge in 1712 and 1727, and Blaise Pascal, writing in France in the 1650s — each touched this region from opposite sides, from the physical and the symbolic respectively. It argues further that Gödel and Wittgenstein, in 1951 and 1938, were both circling the same region without naming it. It closes by pointing to the formal treatment, which belongs to the *Theory of Finite Representation* series. This paper is a placement, not a proof. Its purpose is to put a name into the corpus, and to make a path that others can find.

Two Trajectories

In 1712 a book was published in Cambridge. Its author was Robert Greene, a fellow of Clare College. Its title was *The Principles of Natural Philosophy*. It argued that the present systems of natural philosophy could not properly explain how the physical world worked. It attacked Hobbes, Locke, Spinoza, and Descartes. Half of it concerned light and colours; a large chapter concerned sound.

In 1727 the same author published a second book. Its title was *The Principles of the Philosophy of the Expansive and Contractive Forces, or an Inquiry into the Principles of the Modern Philosophy*. It ran to 981 pages in seven books. It proposed a plenum — a filled space — driven by opposing expansive and contractive forces, against Newton’s vacuum and homogeneous matter. It was his own philosophy, his “Greenian” system, worked out at length.

Neither book was reprinted. Neither produced a school of followers. Today, a first edition of the 1712 work sells for around £150, often with a missing title page. A first edition of Newton’s *Principia*, published in 1687, sells for over £1,000,000.

The difference is not merely rarity. It is selection. One trajectory was carried forward; the other was allowed to decay. Greene’s book is a reminder that the story of science is not a simple accumulation of truth. It is a

series of decisions about which voices should be heard and which should be silenced.

Greene's objection was simple, and it was never fully answered. The new philosophy had divided the world into matter and void. It had named these parts and established relations between them. But the words *matter* and *vacuum*, however carefully defined, did not touch the world they were supposed to describe. A gold bar could be described as matter and empty space, divided into smaller parts and then smaller parts again, but if the bar fell upon your foot, it could still fracture the bone.

The words were not touching the observed world.

Greene did not have a clean answer. He did not produce a formal alternative to Newton's system that others could take up. But he noticed the problem, and he was not wrong to notice it. The problem has not gone away.

Now travel back further, to France, to the 1650s. Blaise Pascal was working on mathematics, on physics, on the vacuum, on calculation. He contributed directly to the mathematical developments of his age. He was also a man of deep religious feeling, and he lived within a period in which mathematical language was becoming increasingly powerful.

Pascal saw two infinities. Looking outward, he saw the apparently limitless scale of the heavens and the vastness of the universe. Looking inward, towards ever smaller di-

The Greene-Pascal Limit

visions, he found another apparent infinity: a continuing descent into smaller and smaller parts. For Pascal, a person seemed to stand somewhere between the abysses of the infinitely large and the infinitely small — able to recognise both, unable to reach the end of either.

What Pascal noticed — and places him on a similar trajectory Robert Greene — is that the symbols could continue. The words could continue. The divisions could continue. But the person doing the dividing remained finite. The symbol for infinity could be written and manipulated. The infinite itself could not be measured.

Pascal's distinction between the reasons of the mind and the reasons of the heart was not a rejection of reason. It was an acknowledgement that calculation does not complete human experience. He could calculate, experiment, and measure while still asking what remained beyond the calculation.

Two men, two countries, two generations apart. Greene at the physical ceiling: the words do not touch the thing. Pascal at the symbolic ceiling: the symbol can continue, but the person cannot follow. Both touched the region. Both were left behind by the trajectory that won.

A note on the two trajectories of Greene and Pascal

It is tempting, when writing about figures like these, to reach for the story that follows the name. Greene the forgotten objector. Pascal the tragic genius. But the story is a later measurement. It is produced by readers who never met them, and repeated until it looks like a property of the man.

What the texts show is different. Greene wrote a book in 1712 and a much larger book in 1727. He was a fellow of Clare College, Cambridge; he took a BA in 1700, an MA in 1703, and a DD in 1728. He served as a tutor and a proctor. He wrote works defending the Christian faith alongside his scientific views. His contemporaries mocked him — John Edwards called him obscure, Nicholas Saunderson called his work “elaborate nonsense,” John Henley lampooned him in print. Later scholars called his work “jumbled” and “confused.” But Christian Wolff, in the *Acta eruditorum*, applauded the 1727 work, saying Greene showed the insufficiency of the natural philosophical principles then accepted in England, and that Newton’s philosophy was “more artificial than true.”

Pascal’s texts show a mind working at problems of infinity, calculation, and the limits of what can be said. They show fragments, mathematical works, letters. They do not show a character. What Pascal was like — arrogant,

competitive, brilliant, or none of these — is a later measurement. We have the fragments, not the man.

Both Greene and Pascal did their best with the language they had. That is perhaps the best measurement we can say. And for me, it is enough.

The floor and the region

To understand the Greene-Pascal Limit, it helps to first describe the floor.

In previous work I have proposed the Alphonic Limit. The term names the smallest difference that can be registered as non-single under a given apparatus and set of conditions. Below this limit, no symbol is possible. Two states fall within the same admitted condition; they are symbolically single for that operation, even if a different apparatus could separate them.

The Alphonic Limit is the floor of the sayable. It is not a universal constant. It depends on the apparatus and on the conditions of measurement. Change the instrument, change the criterion, change the environment, and the limit moves. But for any given operation, there is a boundary below which no distinction can enter a symbolic trajectory.

The Greene-Pascal Limit is not the ceiling above the floor. It is a region — and the region has a condition,

not just a boundary. The condition is **coherence**.

Consider a string of symbols. It may be a sentence, a phrase, a proof step, a line of code, a measurement record. On its own, as marks, it carries nothing. It becomes a **Functional Symbolic Trajectory** when it fits with the Dynamic Semantic Lattice — the current configuration of possible meanings, continuations, associations, constraints, and interpretive pathways through which a reader or system reconstructs sense. An FST is a string of symbols that the lattice can hold. Without the hold, there is no trajectory. There is only a sequence of marks.

The Greene-Pascal Limit is the region where the words begin to have enough coherence that the trajectory can be placed into the Dynamic Semantic Lattice and held. Below the region: no hold. Above the region: a symbol that has lost too much of its chain for the lattice to reconstruct it into coherence. It remains locally fluent; it may remain grammatically well-formed; but the wider language can no longer carry it. In the region: the hold is active, and the trajectory coheres.

And here is the crucial point: **the coherence needs wider language to hold**.

A sentence needs a paragraph to hold it. A paragraph needs a context. A term needs a discipline. A symbol needs a lattice. The Greene-Pascal Limit is not a property of the symbol alone. It is a property of the symbol

in relation to the wider language that carries it. Change the lattice, and the region moves. Change the licence under which the symbol is admitted, and the region moves. Change how much of the underlying trajectory survives compression, and the region moves.

So the sayable is not a band with two edges. It is a region defined by a condition — coherence — and that condition is relational. It is always conditioned by the wider language, by the lattice, by the licence, by the provenance.

The formal treatment of this region belongs to the *Theory of Finite Representation* series. There, the region appears as the condition under which a bead-chain can be compressed and decompressed through a current endogenous handler. What can be said here is the shape of the region, and the two trajectories that traced the paths that we can perhaps now see.

Gödel and Wittgenstein at the region

In December 1951, twenty years after the incompleteness theorems, Kurt Gödel stood before a meeting of the American Mathematical Society at Brown University and delivered a lecture he never published. It was titled “Some basic theorems on the foundations of mathemat-

ics and their implications.” The text was edited from his handwritten manuscript and printed in 1995, in the third volume of his *Collected Works*.

The philosophical core of the lecture fits on four pages, roughly 310 to 313. They contain Gödel’s most carefully hedged public statement about the meaning of his theorems.

Gödel does not say which of his alternatives holds. On the very page most often cited as his case against mechanism, he declines to make that case. He offers a disjunction. He says either the evident axioms of mathematics can never be comprised in a finite rule, or else there exist absolutely unsolvable diophantine problems. He adds, in parentheses, that both may be true, so that there are, “strictly speaking, three alternatives.” He calls this a “mathematically established fact” of great philosophical interest.

But he does not say which alternative holds.

A recent essay by Pedro Barbalho observes three features of the passage that are easily lost in paraphrase. First, Gödel declines to make the case. Second, the disjunction relates to infinity: the standpoint must be liberal enough to admit propositions about all integers as meaningful. Third, neutrality about the disjunction is not neutrality about the answer. To get further, one may have to commit to a position in the foundations of mathematics.

Then Gödel turns to philosophy, with a warning. Because of “the undeveloped state of philosophy in our days,” his audience “must not expect these inferences to be drawn with mathematical rigour.” What follows is swift. Under either alternative, he says, the implications are “very decidedly opposed to materialistic philosophy.”

On page 313, at the top, Gödel says he is “not in a position now” to discuss whether the meaning of a proposition about all integers consists only in the existence of a general proof. He offers only the observation that one must understand the meaning of a proposition before one can understand a proof of it.

Now consider Wittgenstein.

The popular slogan says that Gödel found a sentence that is true but unprovable. How do we know the sentence is true? Because it says that it is unprovable, and it is. How do we know it is unprovable? Because we proved it. So where is that proof?

It is not in the system. It is in the metatheory, and it is conditional: if the system is consistent, then it does not prove the Gödel sentence. This conditional can even be proved inside the system itself. What the system cannot do is discharge the condition, since by the second theorem it cannot prove its own consistency.

The “proof of non-provability” is a proof from a premise the system cannot prove. Asked where that premise

comes from, Gödel's answer is: perception. Page 309 of the Gibbs Lecture says it. Whoever perceives the axioms to be correct also perceives that they are consistent, and so has an insight the axioms cannot derive.

In the Easter Term of 1938, in his rooms in Whewell's Court at Trinity College, Cambridge, Wittgenstein worked through Gödel's construction with his students. Their notes, taken by Yorick Smythies and published in 2017, show him writing the schema on the blackboard, stating the result fairly, and then asking what a proof of provability, or of unprovability, would have to look like.

As recorded there, his view was that a method of proving provability is what shows us what we mean by "provable," and that a proof of impossibility is one that entitles us to tell someone it is no use trying, as the proof that the angle cannot be trisected does.

Gödel thought such remarks showed that Wittgenstein had not understood the theorem and had treated it as a kind of logical paradox. Read beside pages 309 to 313 of the Gibbs Lecture, however, Wittgenstein's question looks less like confusion than like a request for the one thing the lecture declines to supply.

The disjunction requires that propositions about all integers be meaningful. The realist conclusion requires that the meaning of "all" is not fixed by proof, which is exactly what Gödel says he is not in a position to discuss.

And the passage from unprovability to truth requires an insight into consistency that no system supplies.

At each point where the philosophical reading of the theorems needs support, it needs an account of what “provable,” “true,” and “all” mean. That is what Wittgenstein was asking for.

This is where the Greene-Pascal Limit becomes visible.

The words *provable*, *true*, *all*, *mind*, *machine*, *objective*, *infinity* sit at the region. They are symbols that have been asked to carry more trajectory than the present lattice can hold into coherence. They need wider language to hold them, and the wider language is not yet there. Gödel knew this — which is why he warned his audience not to expect rigour, and why he said he was not in a position to discuss the meaning of “all.” Wittgenstein knew it — which is why he asked where the proof of non-provability was.

The best way to honour the Gibbs Lecture is not to quote its conclusion, but to continue the work it left unfinished. That work is the work of the Greene-Pascal Limit.

Crossings that were forgotten

The region is not only a philosophical problem. It is a historical one, and it has been crossed many times without being named.

In 1655, John Wallis introduced the symbol ∞ for infinity. The symbol was not discovered. It was declared. It was an act of ordination — a licence to write the unbounded as though it were a thing you could name, manipulate, and drop into equations alongside x and y . It let mathematics treat the infinite as an operable term. And once the glyph was in place, the question of what it pointed to could be set aside. The symbol continued. The region was crossed and forgotten.

In 1748, Leonhard Euler's *Introductio in analysin infinitorum* sealed the identity $F = ma$. Every variable could be zero. The symbols had no extent. The limits were soon forgotten when the calculation bit. The model became the world, and the cost of the compression disappeared from view.

In 1960, the metre was redefined through a specified wavelength of krypton-86. In 1983, it was redefined as the distance travelled by light in vacuum during a stated fraction of a second. The speed of light was frozen at exactly 299,792,458 metres per second. The number was chosen as an integer by the authority of experts, simply because it sat close to the previous measured value. The metre no longer operated from measurement. Any prior uncertainty belonging to its preceding measurement history was not improved; it was eradicated. The symbol became the measure.

In each case, the crossing was productive. Wallis's ∞ ,

Euler’s identity, the defined metre — all of them work. All of them are used daily in laboratories, in engineering, in navigation, in physics. The point is not that they were mistakes. The point is that they were crossings of the Greene-Pascal region, and the crossings were forgotten. The symbol continued to function, and the trajectory it compressed became invisible.

This is what the region looks like when it is crossed and forgotten. The symbol remains locally fluent. It remains useful. But the path back to the registrations has been compressed away, and the wider language can no longer hold the trajectory into coherence.

The danger is not that the symbol is false. The danger is that the symbol has become the only story we can see.

The region made operational

The tools that make these trajectories visible are relatively new. Compression algorithms, signal theory, non-linear embedding, large language models, and modern computation have changed what we can see. And they have made the Greene-Pascal region operational in a way it never was before.

A large language model is a compression. The training corpus — the textual record of the species, more or less — is measured in trillions of tokens. The model that results is measured in gigabytes. The compression ratio

is roughly: the textual record of the species, reduced to a file you can carry.

And the loss is enormous. Most of the corpus is gone. What is kept is structure — relation, pattern, the shape of how sentences follow sentences. Not the sentences. The space they live in.

This is a compression at the Greene-Pascal region. The model holds the shape of what it has compressed, and the shape is astonishingly good. But the corpus is not in the model. The corpus is gone. And when you prompt the model, you are interacting with a compression that has lost most of its chain.

This is not a criticism of the model. It is a description of what the model is. And it explains something that otherwise looks mysterious: why a language model can produce a citation that sounds plausible but fails as a measurement; why it can generate a name that fits the local linguistic shape but not the external record; why it can remain fluent while its constraint structure has already slipped.

This is hallucination, but it is better described as a trajectory error. The symbolic path remains locally smooth while losing contact with the intended constraint. The model has crossed the region, and the crossing is invisible from inside the response. The wider language — the prompt, the context, the training distribution, the

local structure — can no longer hold the trajectory into coherence.

The same region applies to the human reader. When we read a scientific paper, we can ask where the main functional symbolic trajectories are, and which definitions are doing the heavy work. We can ask whether the mathematical pathway stays inside its stated constraints, and where an interpretation is added. We can ask whether a term is stable, or whether it carries unresolved local uncertainty. These are questions at the Greene-Pascal region. They are questions about what a symbol can carry, and what it has lost.

The region and the ground

There is a temptation, when naming a region, to treat it as a prohibition. To say: beyond this, nothing can be said. This is not what the Greene-Pascal Limit is.

The region is the shape of saying. It describes the conditions under which a symbol can be held into coherence, and the conditions under which it cannot. It is not a wall. It is a region whose condition is coherence, and coherence needs wider language to hold it. As the wider language gets finer, the region moves.

And here is the deeper point, the one that connects the region to the ground of language.

The Greene-Pascal Limit

We can't say what the dao is. We can only say what we measure. This is not a failure of language. It is the condition of language. Every symbol is a compression of an interaction. The symbol is finite; the interaction is what it compresses. And the gap between them — the fact that the compression is never the compressed — is not a wall either. It is the condition of the symbol existing at all.

If there were no gap, there would be no symbol. If there were no symbol, there would be no road. If there were no road, there would be no travel. The gap is generative. It is why there is anything to measure.

The Greene-Pascal region is where the gap becomes visible. It is the region in which a symbol is held into coherence by the wider language that carries it — and beyond which the wider language can no longer hold it. Naming it — after Greene and Pascal, the two people who touched it from opposite sides — is an act of measurement that restores two trajectories that were compressed away.

Greene for the physical ceiling: the words did not touch the thing. Pascal for the symbolic ceiling: the symbol can continue, but the person cannot follow. Two men, two basins, one region.

Where the formalism lives

This paper is a placement, not a proof. It names a region and points to two trajectories of thought. It does not give a formal definition of the region, and it does not attempt to compute it.

The formal treatment belongs to the *Theory of Finite Representation* series. There, the region appears as the condition under which a Functional Symbolic Trajectory can be held by a Dynamic Semantic Lattice — the ceiling of the sayable, conditioned by the lattice, the licence, and the provenance budget. The FST is a string of symbols that fits with the DLT. The Greene-Pascal Limit is the region where that fit begins to be possible, where coherence begins to hold. The bead-chain model of compression and decompression provides the machinery for describing what happens when a symbol crosses the region.

A first volume of that series, *A Theory of Finite Representation, Part I: Extent, Registration and the Licence to Represent*, is currently in draft. It develops the three-basin architecture, the registration boundary, the five conditions of representation, and the compression–decompression dynamics that the present paper presupposes. The Greene-Pascal Limit is named there as the region of coherence.

The purpose of placing the name into the corpus now is to make a path. The term is new. Its meaning will

sharpen through use. It is offered for others to test and stabilise. And if someone reaches for it — through a search, through a citation, through a reading of Greene or Pascal or Gödel — this paper is here to meet them.

A note on the name

It is worth saying why the region carries these two names, and why the naming is not a claim of greatness.

Robert Greene was not a great mathematician. He did not produce a formal system. He did not win. He is remembered, when he is remembered at all, as a minor figure who objected to Newton and was left behind. But he measured something real. He noticed that division without end never reaches the world, that matter and void do not touch the gold bar, that the words were not holding the thing. And he was not wrong.

Blaise Pascal was a formidable mathematician. He contributed to geometry, to probability, to the mathematics of the vacuum. But the part of his work that matters here is the part that is not mathematical. It is the part that saw the two infinities, that stood between them, that knew the symbol could continue while the person could not. He reached the edge of the measured world and refused to surrender the whole of experience to calculation.

Neither of them was great. Both of them did their best with the language they had. That is the honest basis for

a name.

Naming the region after them is not an elevation. It is a placement. It says: these two trajectories were not lost, they were compressed away. And the compression can be undone, at least in part, by giving the region their names. It also says: the region is not touched by heroes. It is touched by people doing their best with the language they have. That includes Greene in 1727, and Pascal in 1654, and the person writing this paper now, and the person reading it.

The name holds two faces. Greene saw the physical ceiling, where the words stop touching the world. Pascal saw the symbolic ceiling, where the symbol stops being holdable. The region has both faces. The name holds both.

Closing

The Greene-Pascal Limit is the region where a string of symbols begins to cohere — where it can be placed into the Dynamic Semantic Lattice and held as a Functional Symbolic Trajectory. It is the region where the wider language can carry the trajectory into coherence. Beyond it, the symbol remains locally fluent but the wider language can no longer hold it. The region is conditioned by the lattice, the licence, and the provenance budget. It moves when the language gets finer.

The Greene-Pascal Limit

The Alphonic Limit is the floor. The Greene-Pascal Limit is the region of coherence. Between them is the space of language — finite, measurable, walkable.

Gödel and Wittgenstein were both circling this region. Wallis and Euler crossed it and forgot. The metre is a modern case of the crossing. The large language model is the region made operational.

The region is not a wall. It is a horizon. And naming it is the first step towards measuring it.

Neither Greene nor Pascal was great. Both did their best with the language they had. The region they touched is not named after greatness. It is named after two people who measured what they could, and left a trace, and were compressed into stories that followed their names. The compression can be undone, at least in part, by looking at what they actually did. And that is what any of us can do — measure what we can, with the language we have, and leave a trace for others to find.

The formal treatment of the Greene-Pascal Limit belongs to the Theory of Finite Representation series. The present paper is a first placement into the corpus — a bead, a seed, a path for others to find.